

From Pets To Cattle To Agents: Evolving Identity And Security For Workloads



https://tinyurl.com/pets-sreday

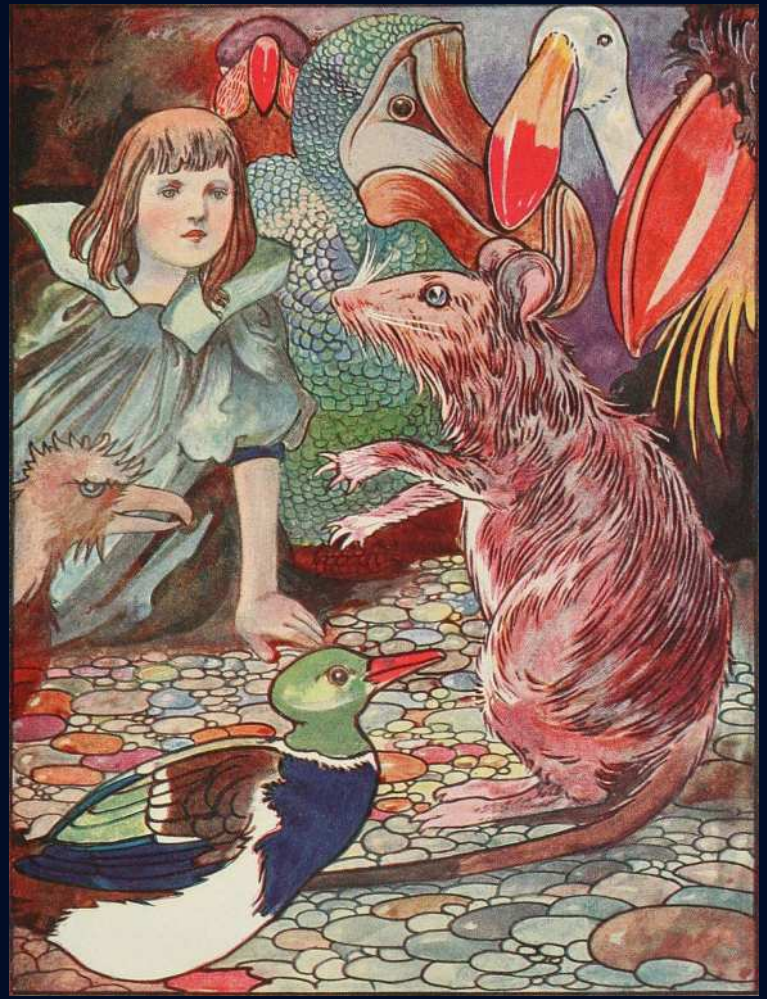




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Anthropomorphism

From Wikipedia, the free encyclopedia

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This article is about the attribution of human traits, emotions, or intentions to non-human entities. For the representation of a thing or abstraction as a person, see Personification.

Anthropomorphism is the ascribing of human personality, appearance, conduct, cognition, or other attributes to non-human entities, often including non-human animals.[1] In fiction and folklore, it is specifically the endowing of non-human characters with human-like behaviors, speech, facial expressions, etc; common examples include intelligent talking animals, talking trees, anthropomorphized food, and sentient toys.

As a general human tendency, anthropomorphism is considered innate to human psychology.[2] Personification, which usually refers to a literary device, is the related attribution of human form and characteristics to non-animal organisms, inanimate objects, or abstract concepts such as nations, emotions, and natural forces like weather or the seasons. Both have ancient roots as storytelling and artistic devices, and most cultures have traditional fables with anthropomorphized animals as characters. People have also routinely attributed human emotions and behavioral traits to wild as well as domesticated animals.[3]



Mickey Mouse, an anthropomorphic mouse and an American cartoon character co-created in 1928 by Walt Disney and Ub Iwerks, showing typical elements of anthropomorphism: bipedalism, human-like hands with opposable thumbs, human facial expressions, and wearing clothing.

Etymology

Anthropomorphism and **anthropomorphization** derive from the verb form *anthropomorphize*,[a] itself derived from the Greek *ánthrōpos* (ἄνθρωπος, lit. "human") and *morphē* (μορφή, "form"). It is first attested in 1753, originally in reference to the heresy of applying a human form to the Christian God.[b][1]

Examples in prehistory

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Anthropomorphism is the ascribing of human personality, appearance, conduct, cognition, or other attributes to non-human entities

As a general human tendency, anthropomorphism is considered innate to human psychology.

We are not talking about “Personification,” which usually refers to a literary device, is the related attribution of human form and characteristics to non-animal organisms, inanimate objects, or abstract concepts such as nations, emotions, and natural forces like weather or the seasons.



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YouTube Search Create

What “toxic anthropomorphism” means — and why it matters



Dangerous Data: Why Meaning Is the New Cyber Battleground | Wendy Nather

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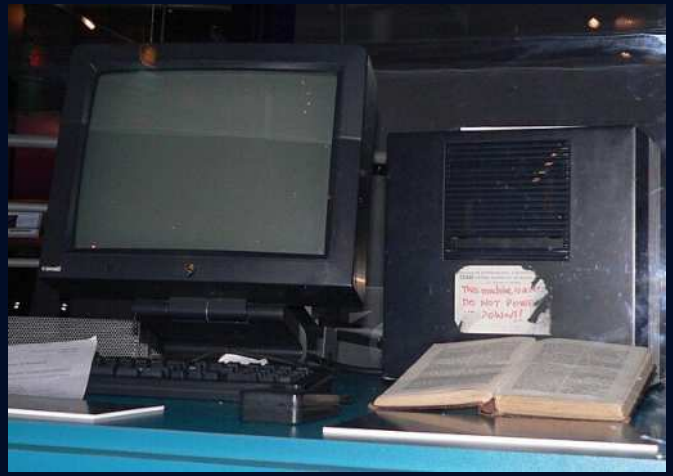




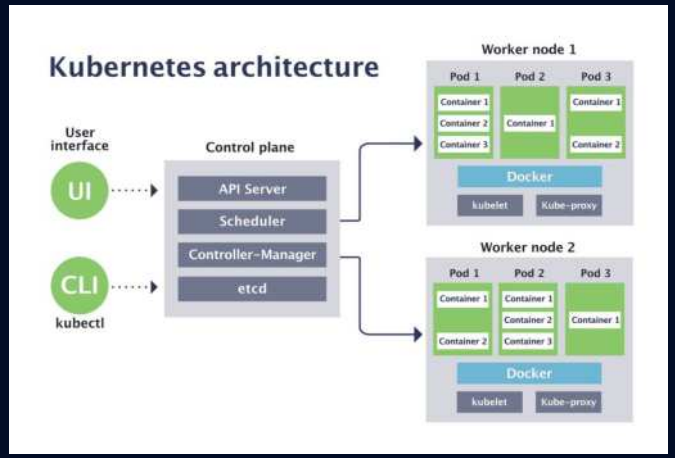




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Hi. I'm Dwayne :



Dwayne McDaniel
Principal Developer Advocate
dwayne.mcdaniel@gitguardian.com

{

"Hometown" : "Chicago",

- "Mission" : "Help people figure stuff out",
- "Developer-advocate-since" : "2014",
- ~~"Host" : "The Security Repo Podcast",~~

"Co-Organizer" : "AppSec Village",

"Website" : "https://Dwayne-McDaniel.com",

- "Socials" : {

"mcdwayne@mastodon.social",

"https://ww.linkedin.com/in/dwaynemcdaniel" },

"Other-interests" : {"crochet", "karaoke",

"rock and roll concerts", "music in general"}

}

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About GitGuardian



GitGuardian is a secrets visibility platform helping teams solve Non-Human Identity security crisis

A Brief History Of DevOps



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Security implemented:
- Lock the door
- Put a guard out front

Prevents:
- Stealing paper punch cards
- Messing with the hardware



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Security implemented:
- Lock the door
- Username/Passwords

Prevents:
- Students from playing Zork using another user's account
- Sending emails from accounts you don't own
- Read "The Cuckoo's Egg" by Clifford Stoll

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Operations

Mission: Keep Server Alive



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Operations

Mission: Keep Server Alive

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PROPRIETE CERN

Manifesto for Agile Software Development

We are uncovering better ways of developing
software by doing it and helping others do it.
Through this work we have come to value:

Individuals and interactions over processes and tools
Working software over comprehensive documentation
Customer collaboration over contract negotiation
Responding to change over following a plan

That is, while there is value in the items on
the right, we value the items on the left more.



DEV





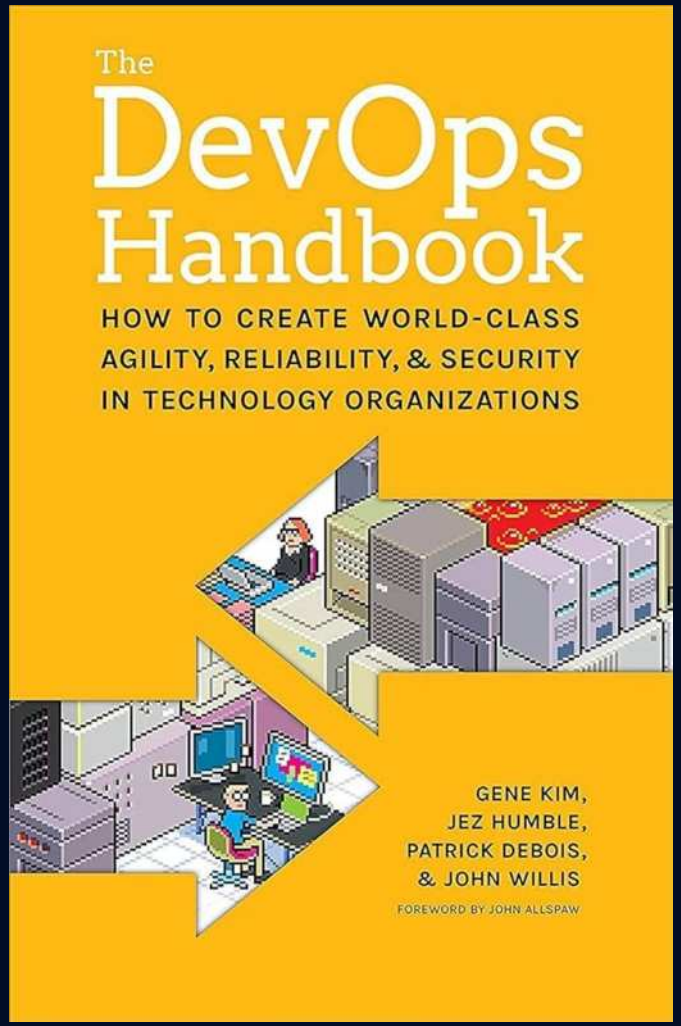
OPS

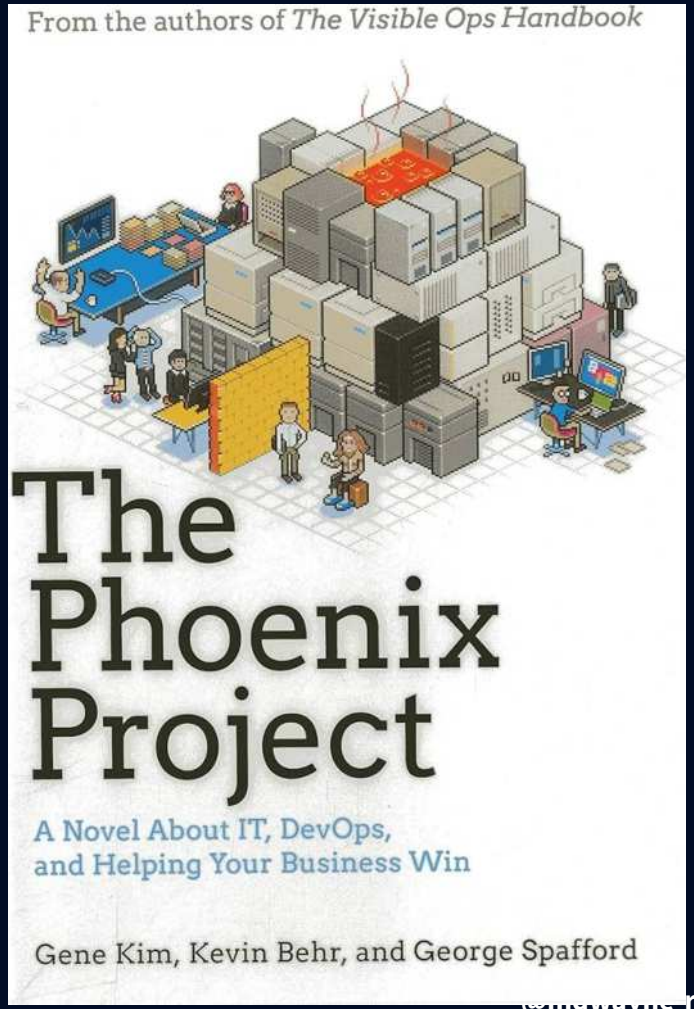
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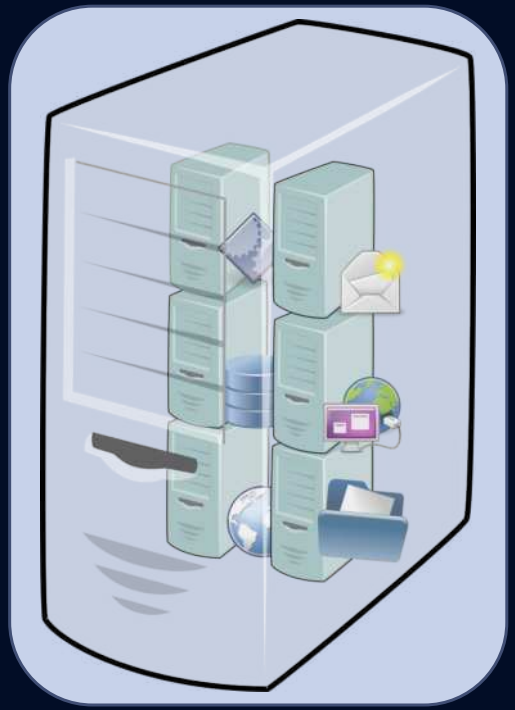
DEV OPS

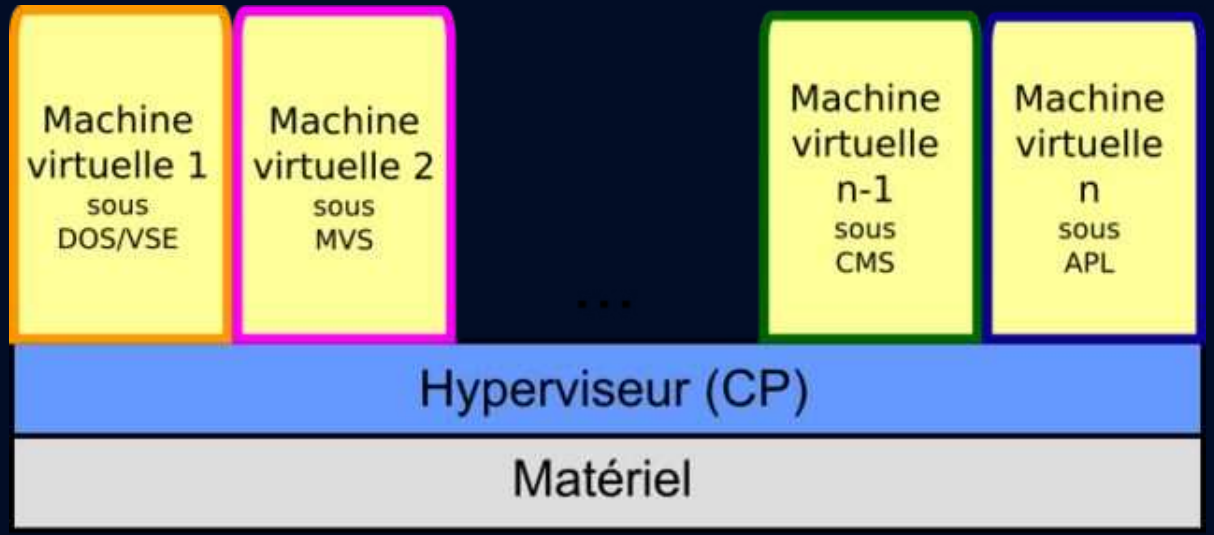
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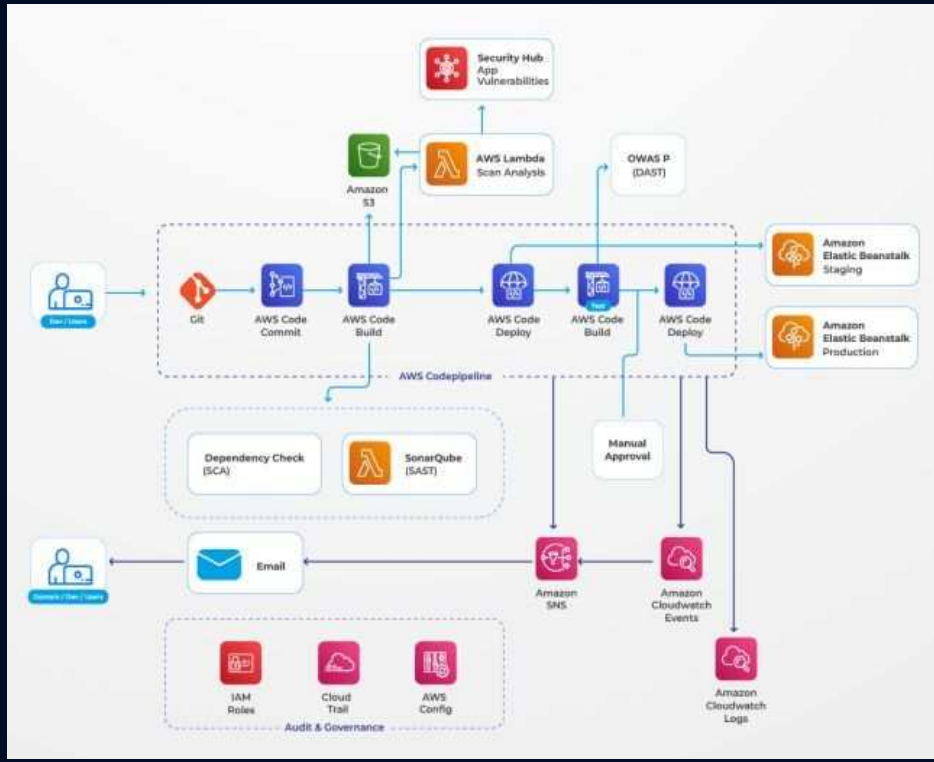
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Security implemented:
- API keys
- SSH keys
- Certificate Authorities (best case)

Prevents:
- Unauthorized access of your machine resources
- Unauthorized access to your admin accounts



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~~Declared Infrastructure State Configurations Stored As Code~~

Infrastructure as Code



HashiCorp Terraform



Pulumi



AWS CloudFormation

OpenTofu



Crossplane





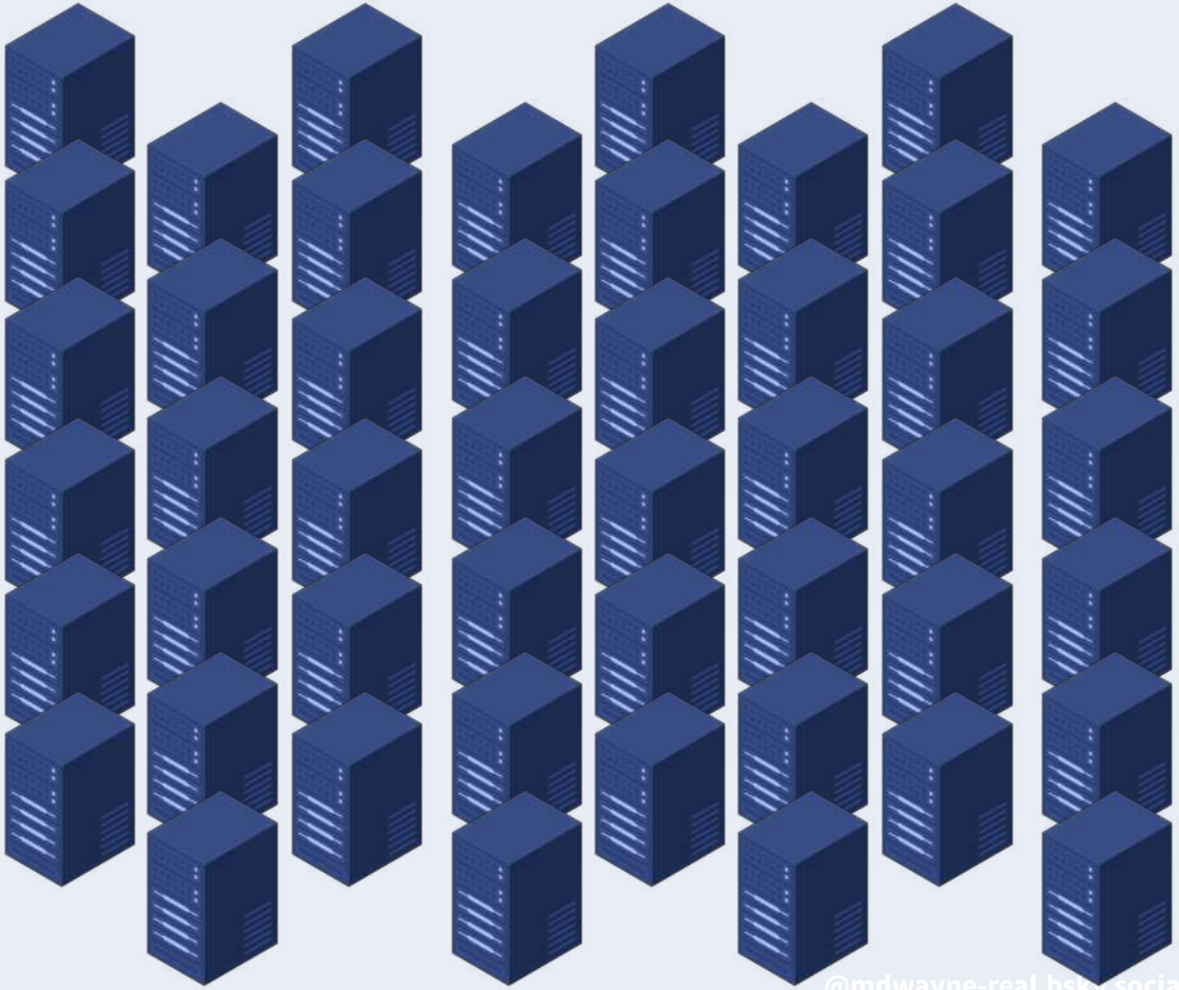


Number_of_Servers = 1

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Number_of_Servers = 48



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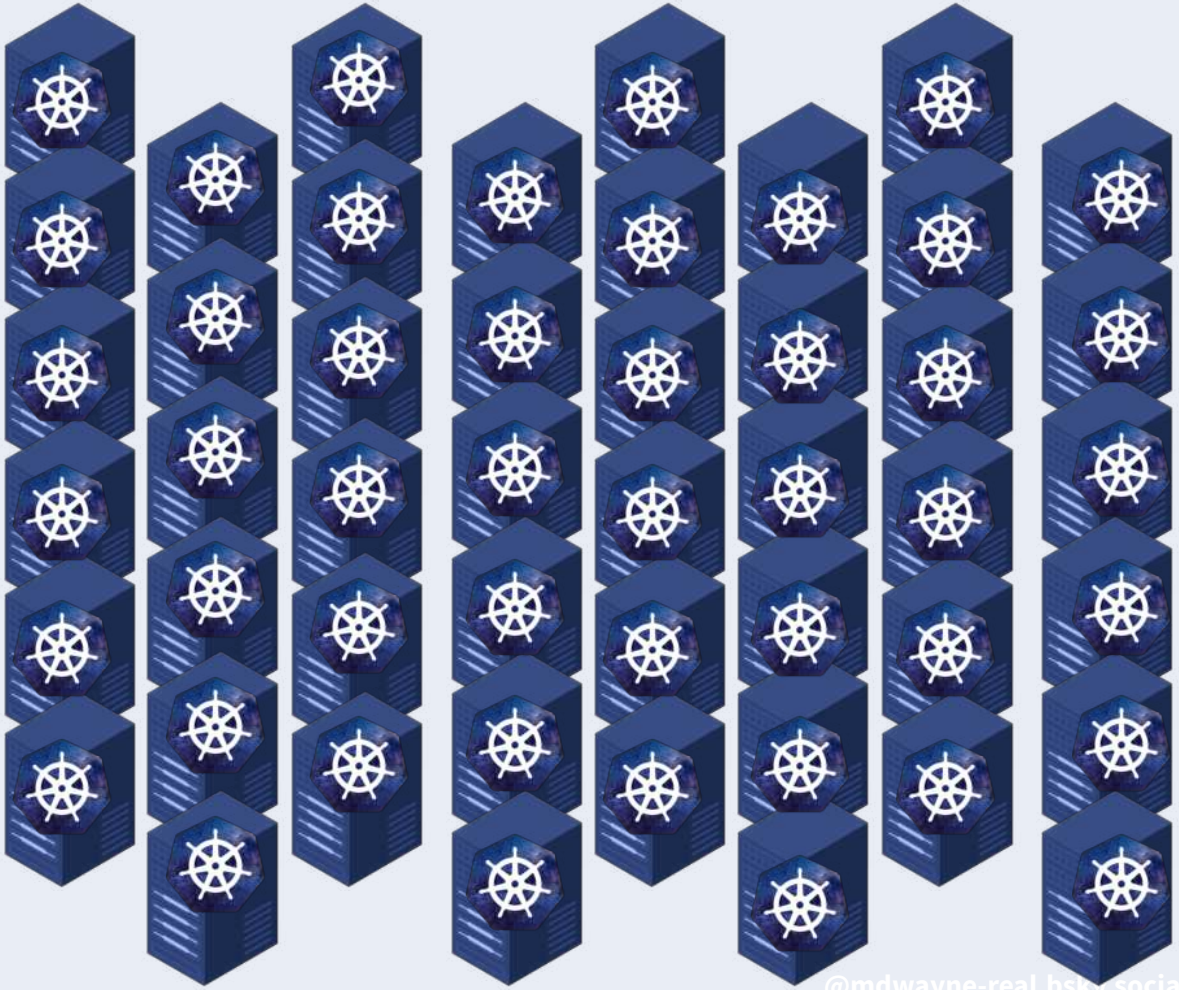


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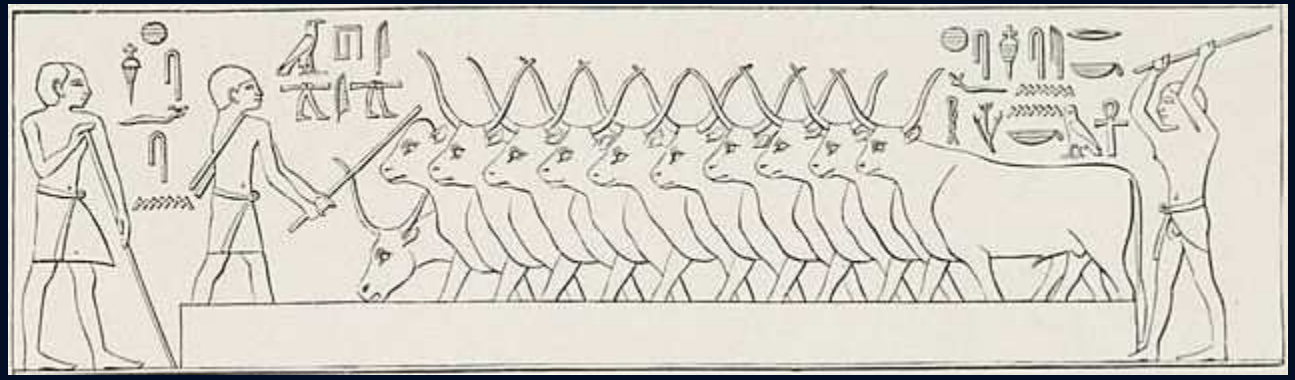
```
securityContext:
  runAsNonRoot: false
```



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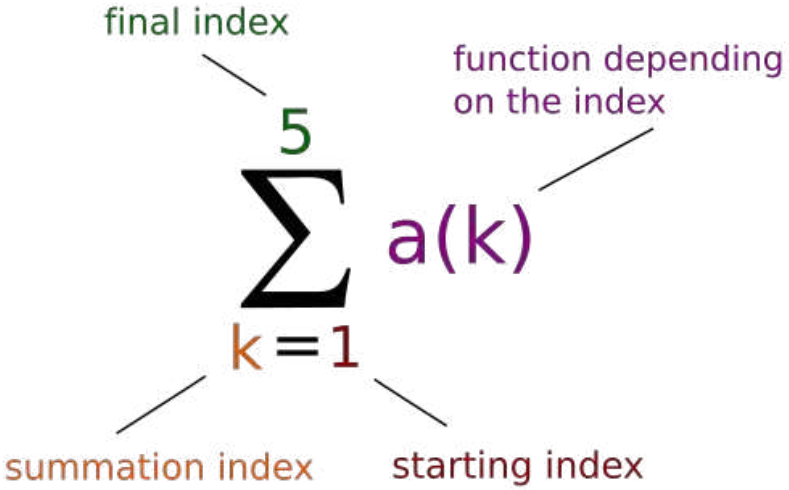




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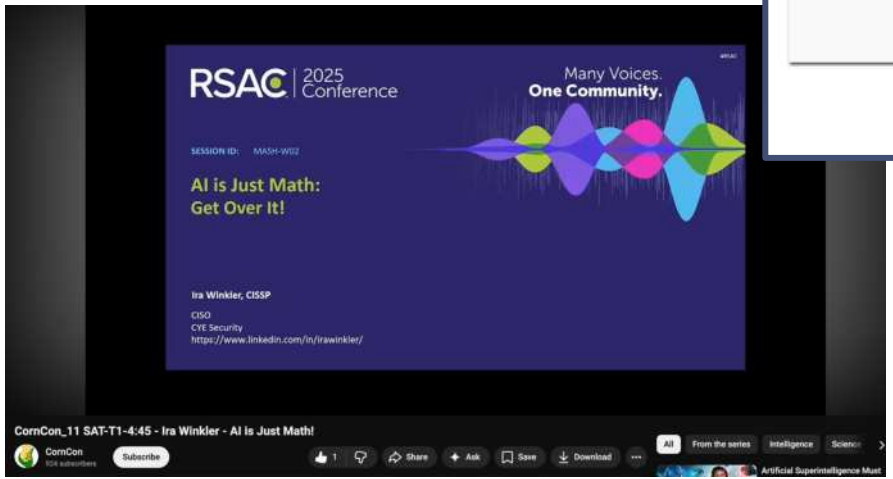
Matrix Factorization for Explicit Feedback

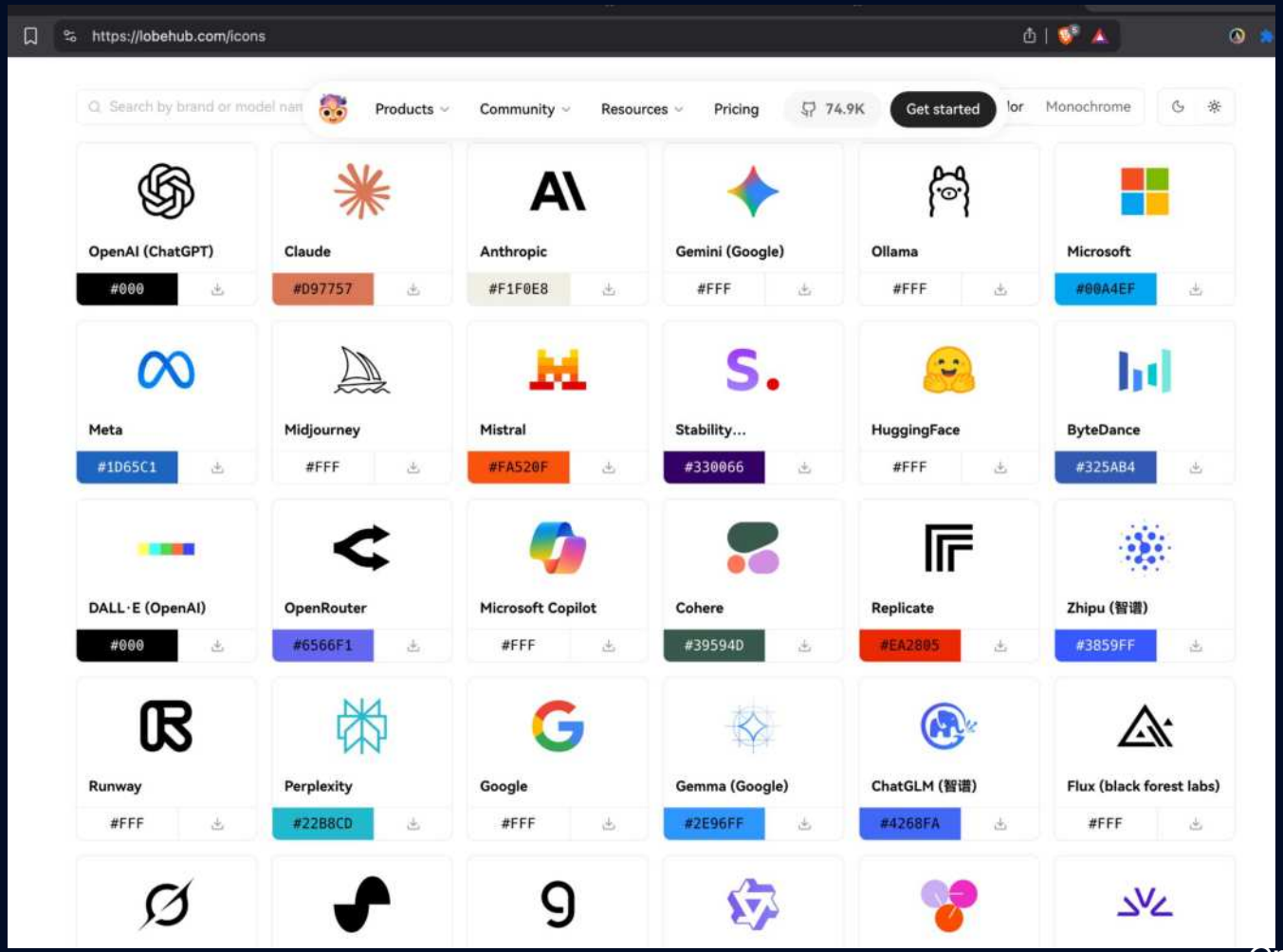
The simplest idea is to model user-item interactions through a **linear model**. To learn the values of p_u and q_i, we can minimize a regularized MSE loss over the set K of pairs (u, i) for which r_{ui} is known. The algorithm so obtained is called **probabilistic matrix factorization (PMF)**.

$$\hat{r}_{ui} = q_i^T p_u$$

$$L = \sum_{(u,i) \in K} \left[(r_{ui} - \hat{r}_{ui})^2 + \lambda \left(\|p_u\|_2^2 + \|q_i\|_2^2 \right) \right]$$

Probabilistic matrix factorization: model for r_{ui} *and loss function.*





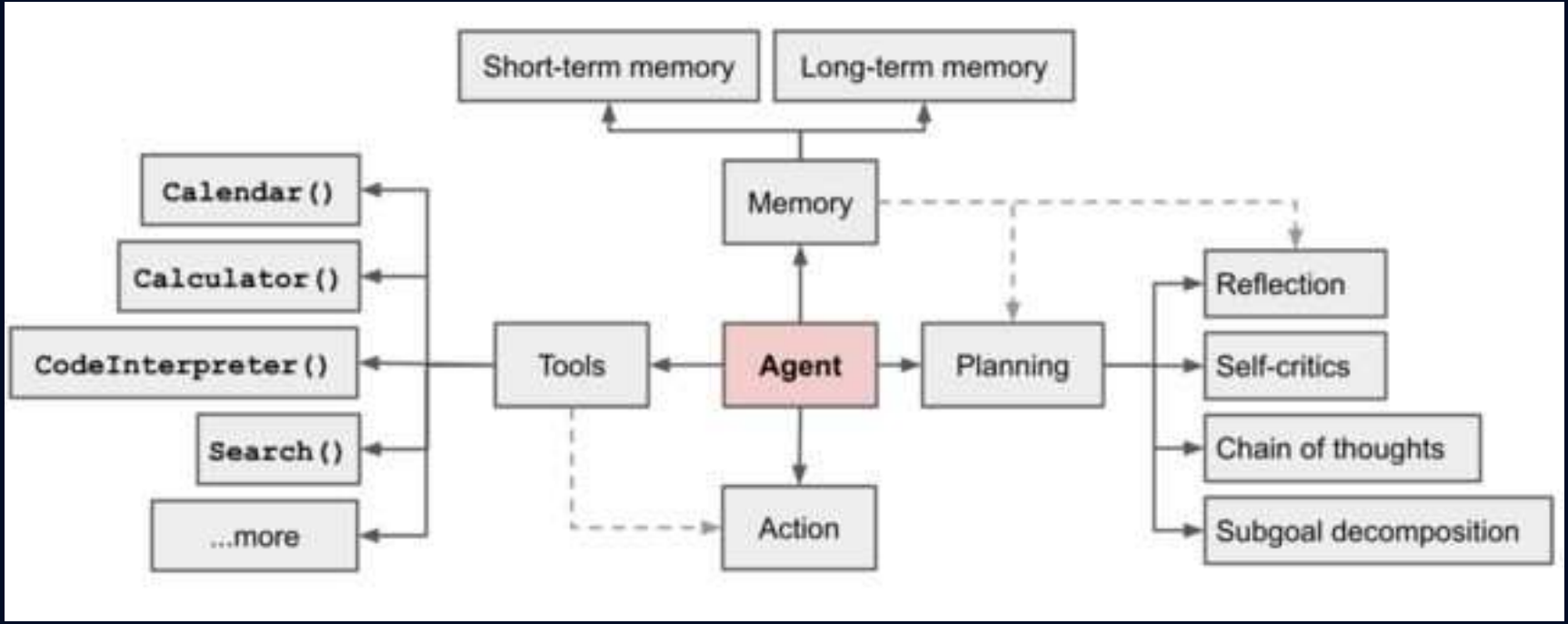
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Agentic AI:

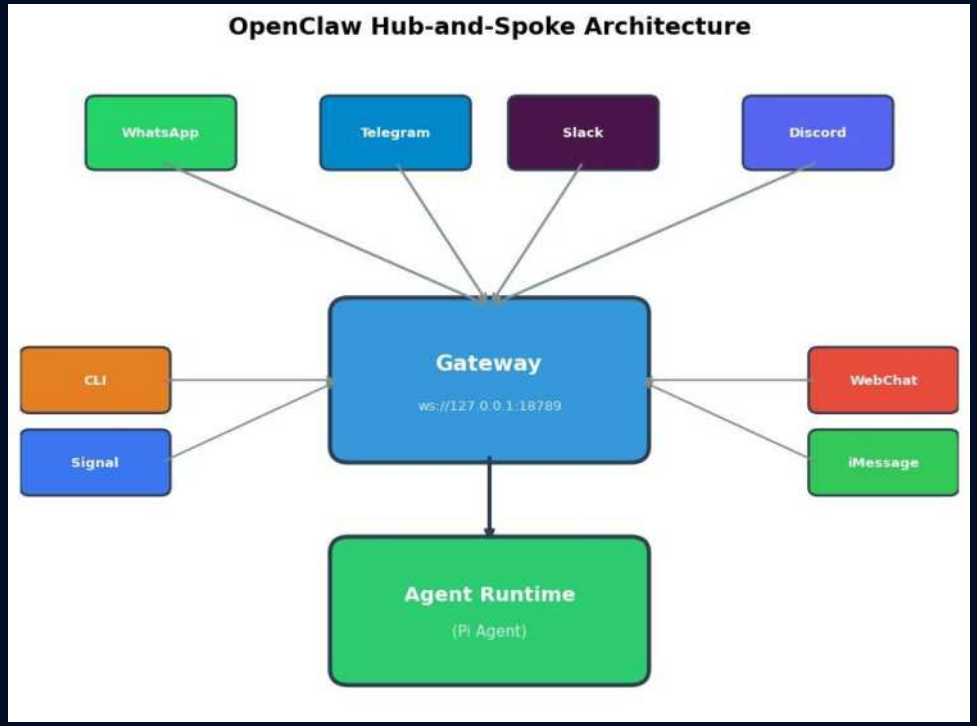
AI systems that can pursue goals by taking actions on its own, using feedback from its environment to decide what to do next.

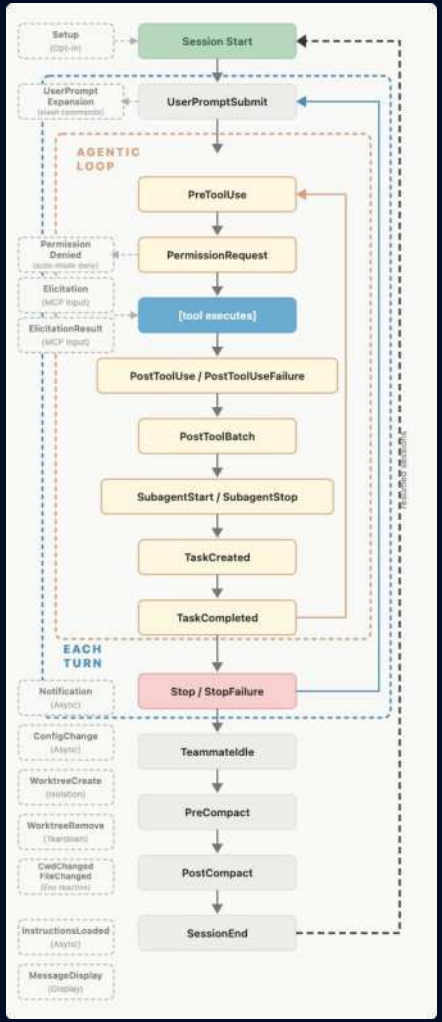


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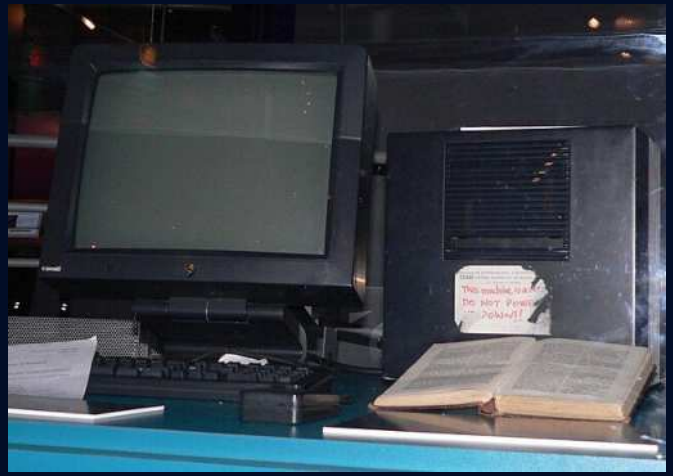


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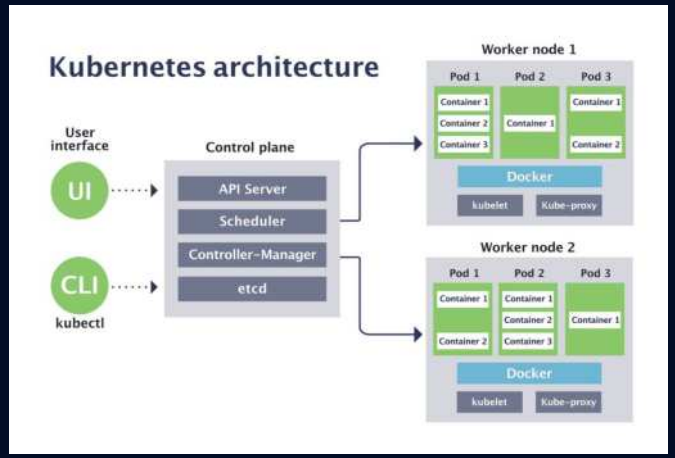




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Human

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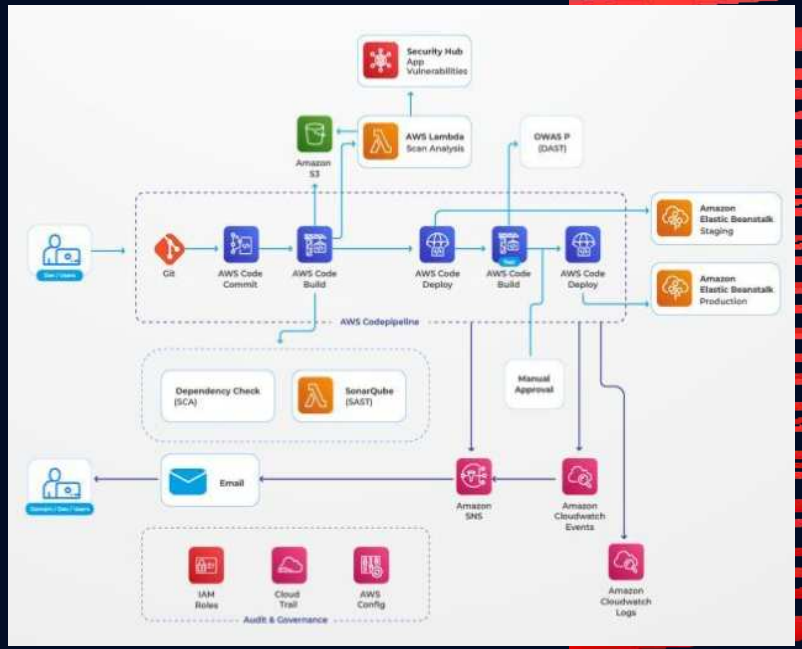
!=



Human

Non-human

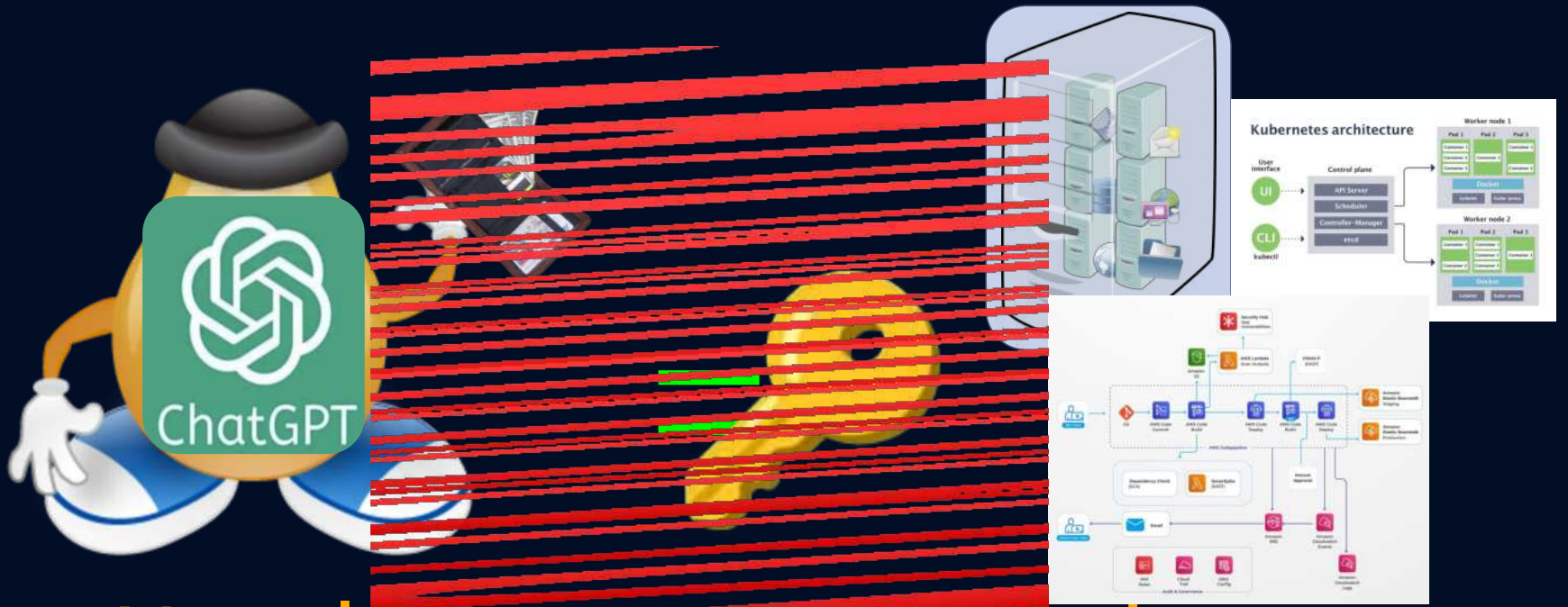
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Non-human

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Non-human Non-human

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What are secrets in software development?

```python
from typing import Dict
import aws_lib

def aws_upload(data: Dict):
    database = aws_lib.connect("AKIAF6BAFJKR45SAWSZ5",
    "hjshnk5ex5u34565AWS654/JKGjhz545d89sjkja")
    database.push(data)
```

Definition

Secrets authenticate access and encryption of software components, such as:

- API keys
- Username/password pairs
- Database connection URLs
- Browser session tokens
- Certificates

Sensitive files such as **.env, .pem or .crt** are also secrets themselves



GitGuardian

THE STATE OF

Secrets Sprawl 2026

www.gitguardian.com

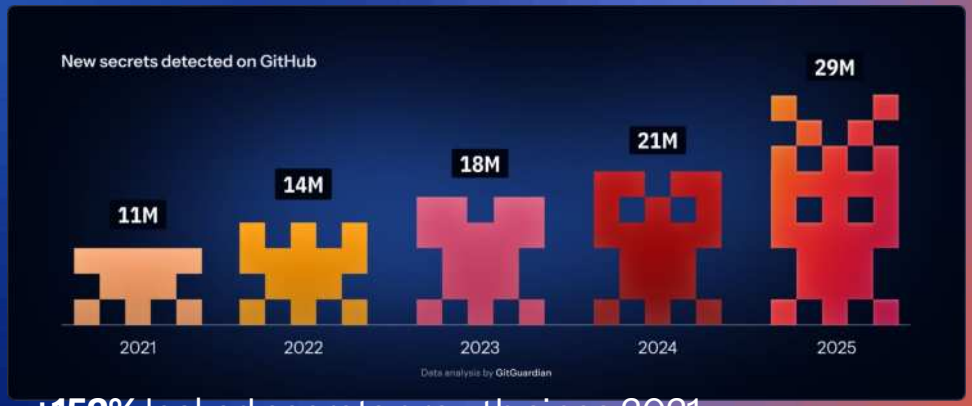
https://www.gitguardian.com/state-of-secrets-sprawl-report-2026

https://www.gitguardian.com/state-of-secrets-sprawl-report-2026

28,649,024

34%

New secrets detected in public GitHub commits in 2025.



+152% leaked secrets growth since 2021
while the developer population only grew 98%

1.94 billion public GitHub commits in 2025

5.6% of public repositories contained at least one hardcoded secret

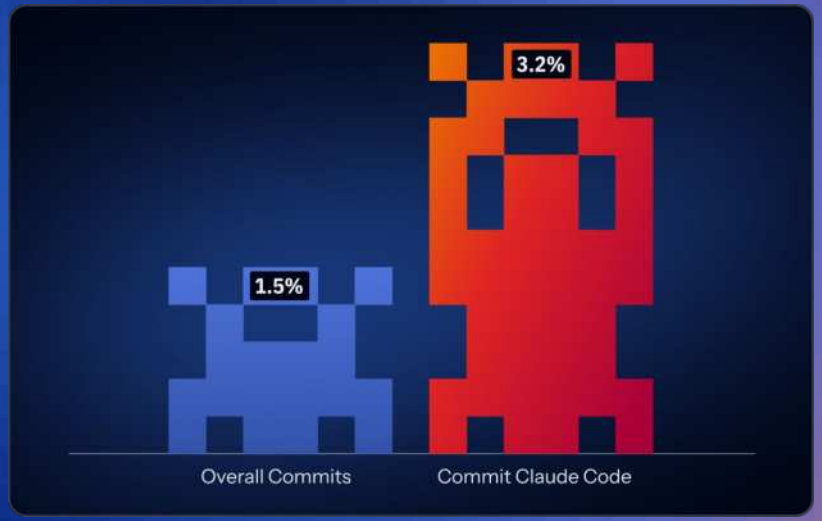
1.5 secrets per 1,000 commits

64% of valid secrets from 2022 are still active and exploitable

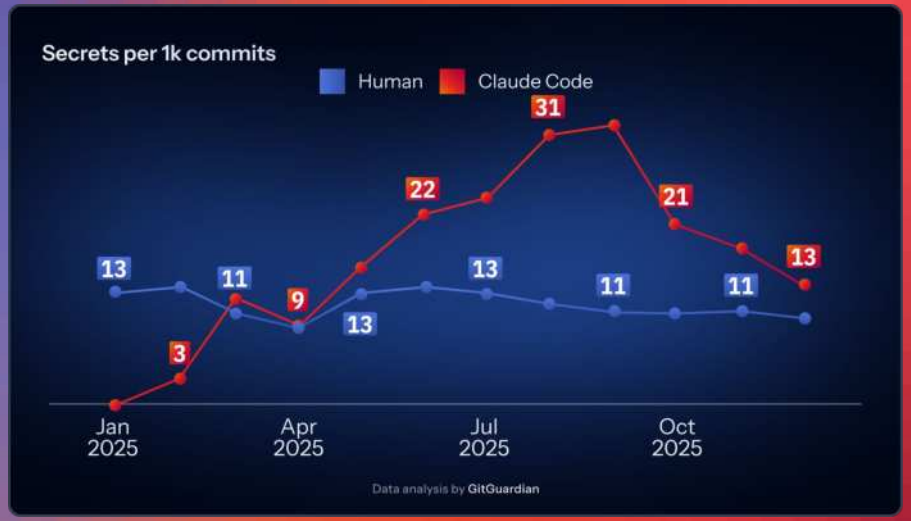
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A look into the results from 2026 State of Secrets Sprawl

Claude Code Impact on Leaked Secrets

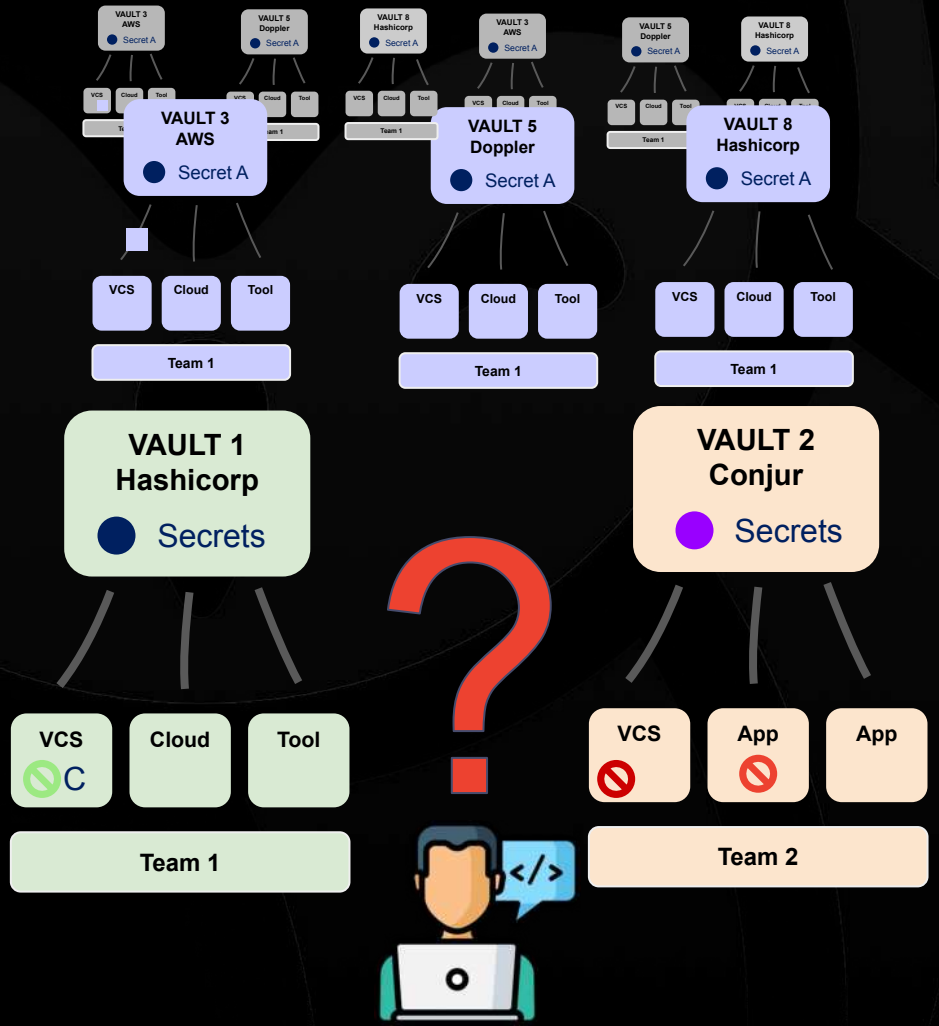


→ These commits climbed to a peak of 31 secrets per 1,000 commits in August. 2.4x the human baseline



→ **Claude Code** co-authored commits **leak secrets at ~2×** the baseline across all Public GitHub commits

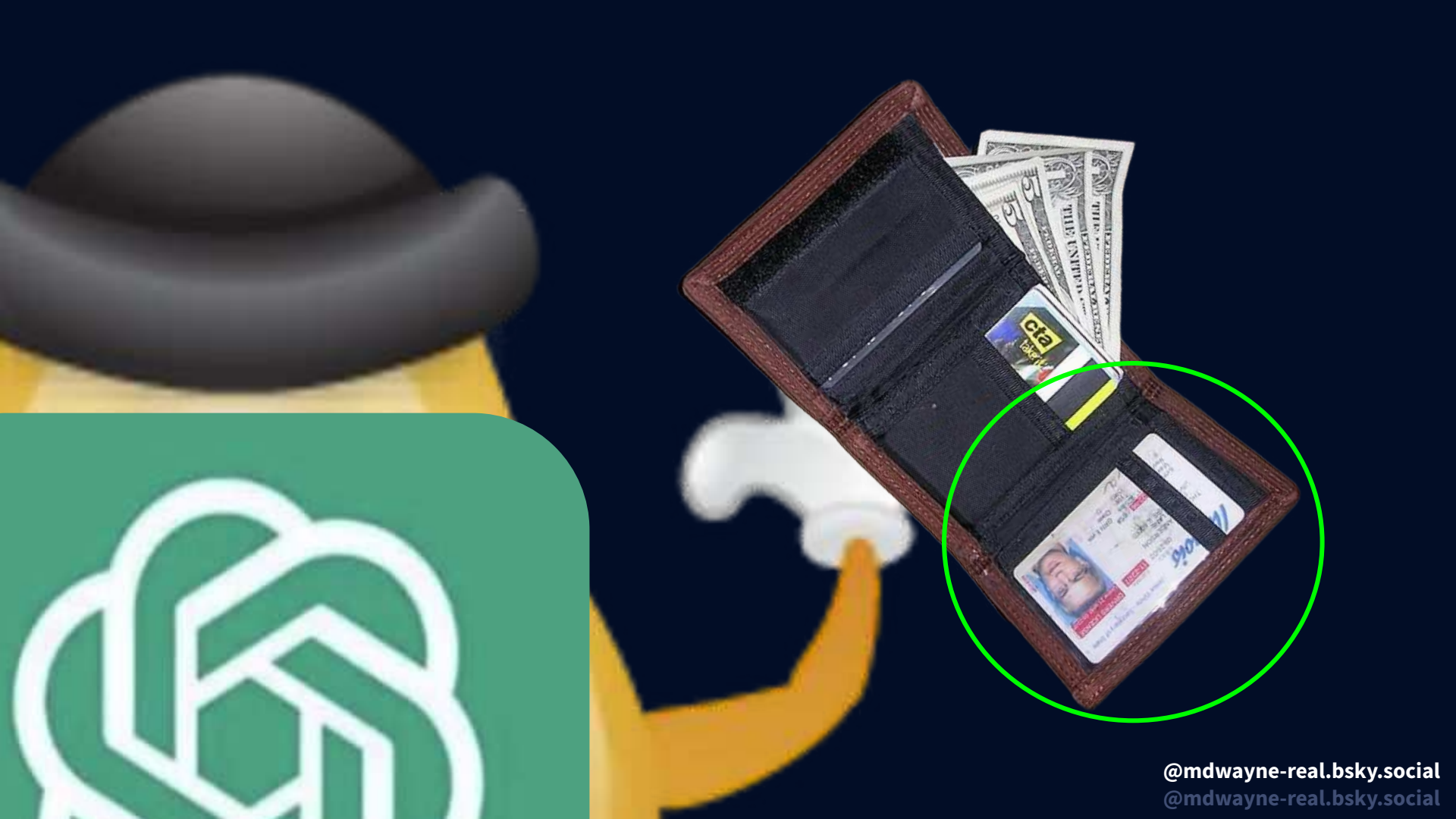
Vault Sprawl



What Can We Do?



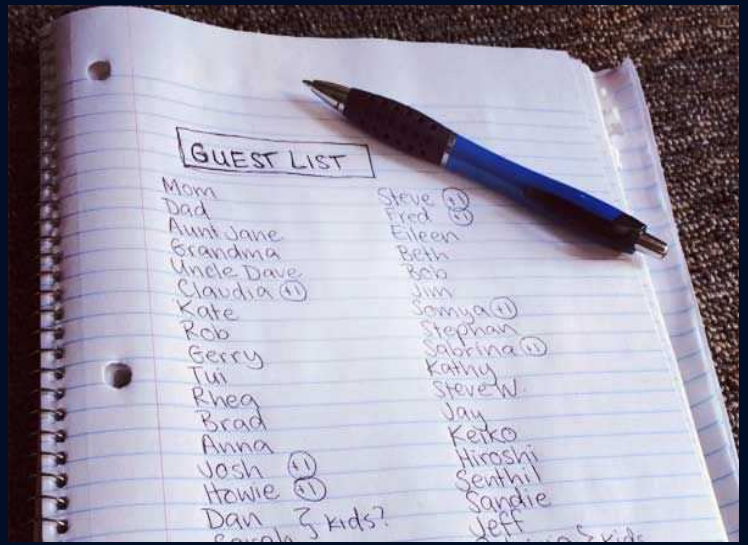
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Paths Ahead For Workload Based Authentication:

1. SPIFFE/SPIRE
2. Istio and Envoy
3. Security Token Services (STS)
4. WIMSE and Workload Identity Tokens (WIT)
5. Federated Identities on Cloud Providers



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Universal identity control plane for distributed systems

SPIFFE and SPIRE provide strongly attested, cryptographic identities to workloads across a wide variety of platforms

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SPIFFE, the Secure Production Identity Framework For Everyone (SPIFFE) Project defines a framework and set of standards for identifying and securing communications between application services.

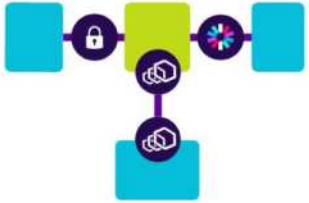


SPIRE (the SPIFFE Runtime Environment) is a toolchain of APIs for establishing trust between software systems across a wide variety of hosting platforms.



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Key Use Cases



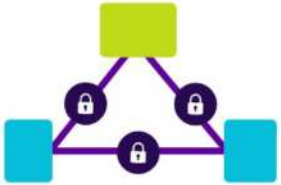
Secure microservices communication automatically with Envoy, X.509 PKI, or JWT



Authenticate securely to common databases or platforms without passwords or API keys



Build, bridge, and extend service mesh across organizations without sharing keys

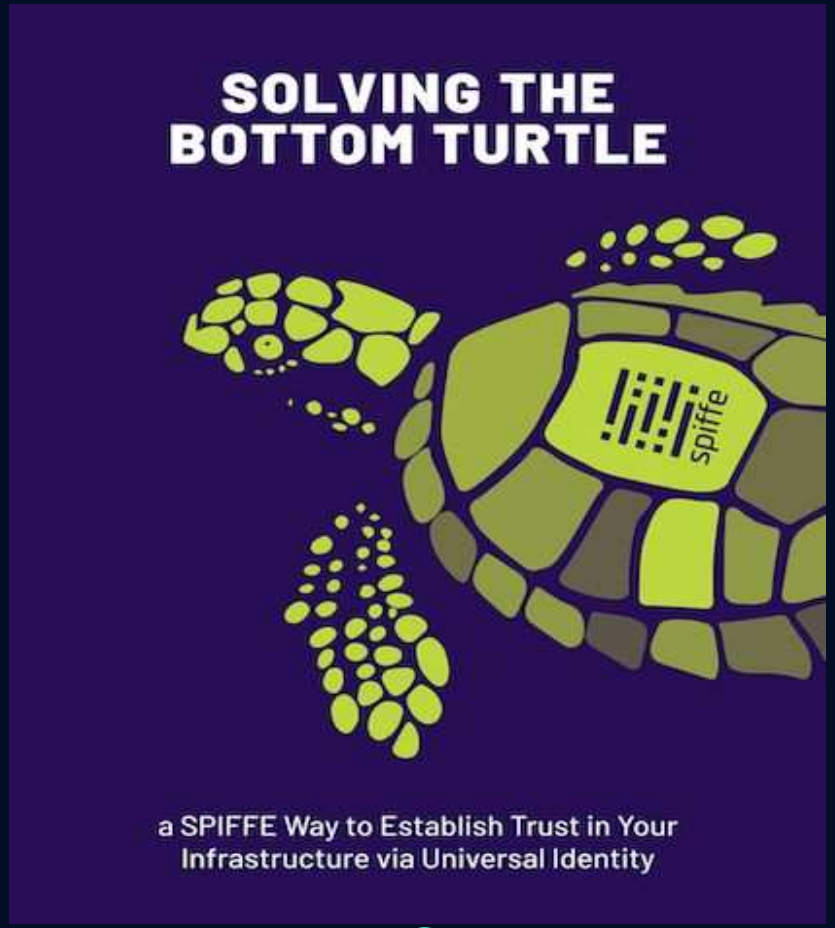


Cross-service authentication for zero trust security model



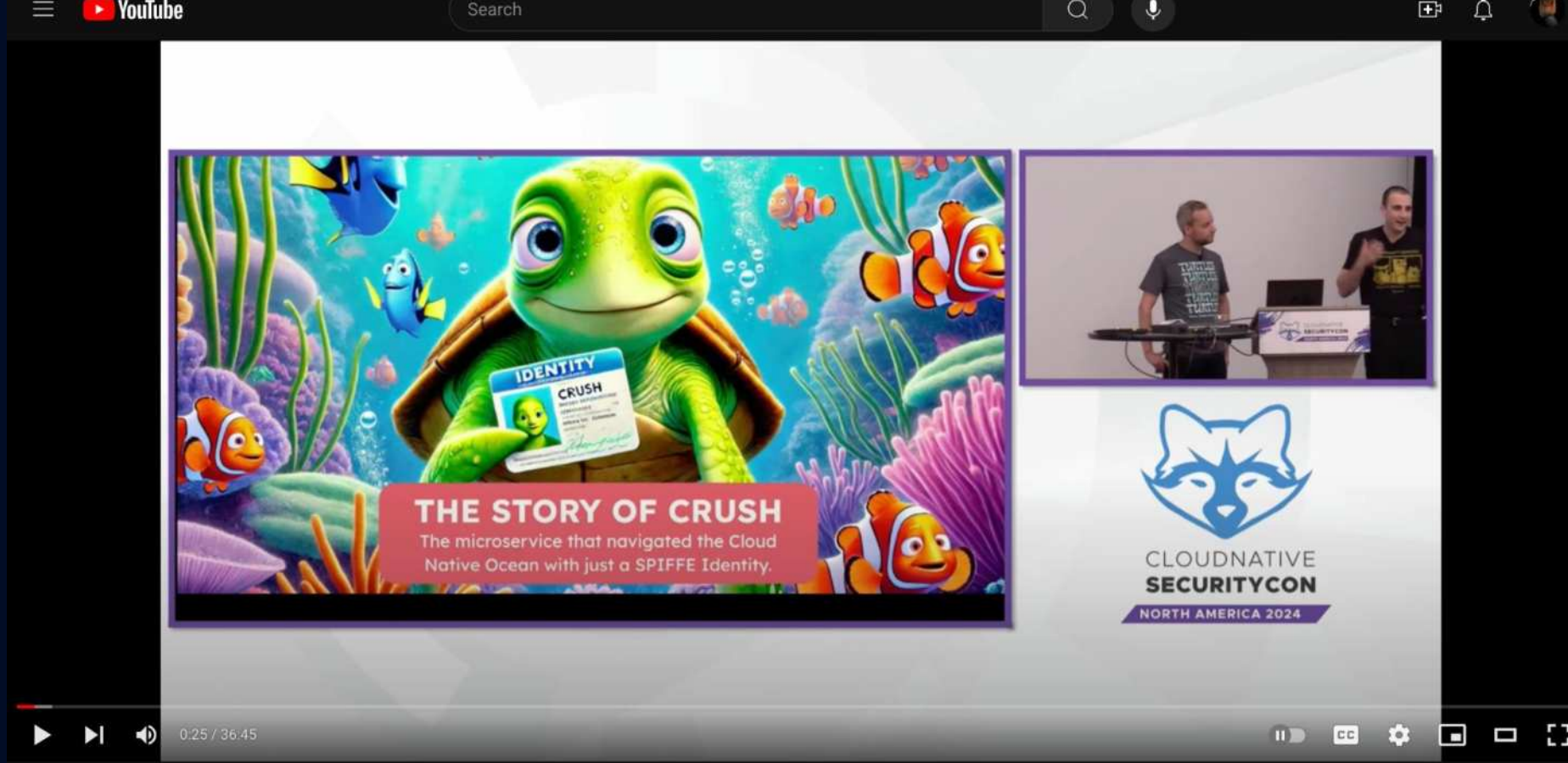
Bridging the gap between Kubernetes and other platforms

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https://spiffe.io/book/

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https://youtu.be/4HHvEamsxjs?si=KZhaCL_51QKAMuG3

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Kubernetes v1.33: From Secrets to Service Accounts: Kubernetes Image Pulls Evolved

By **Anish Ramasekar (Microsoft)** | Wednesday, May 07, 2025



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How it works

1. Service Account tokens for credential providers

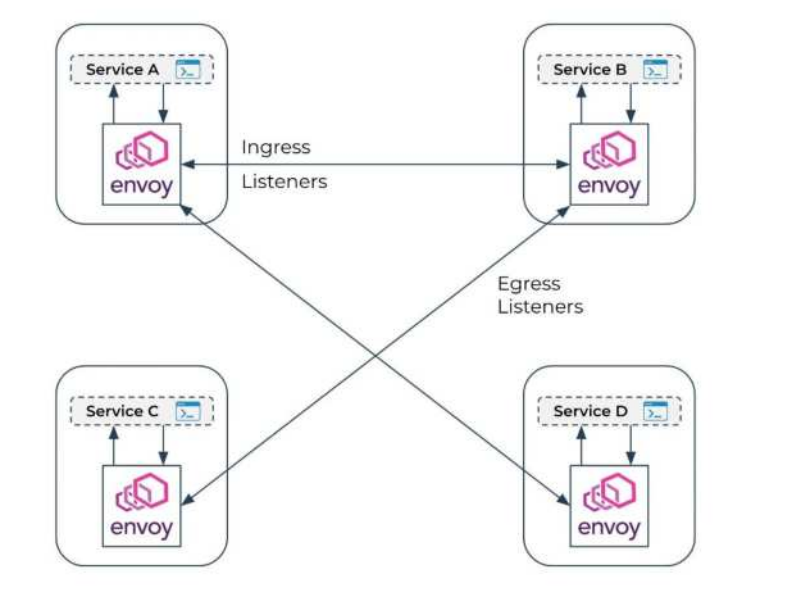
Kubelet generates **short-lived, automatically rotated** tokens for service accounts if the credential provider it communicates with has opted into receiving a service account token for image pulls. These tokens conform to OIDC ID token semantics and are provided to the credential provider as part of the `CredentialProviderRequest`. The credential provider can then use this token to authenticate with an external service.



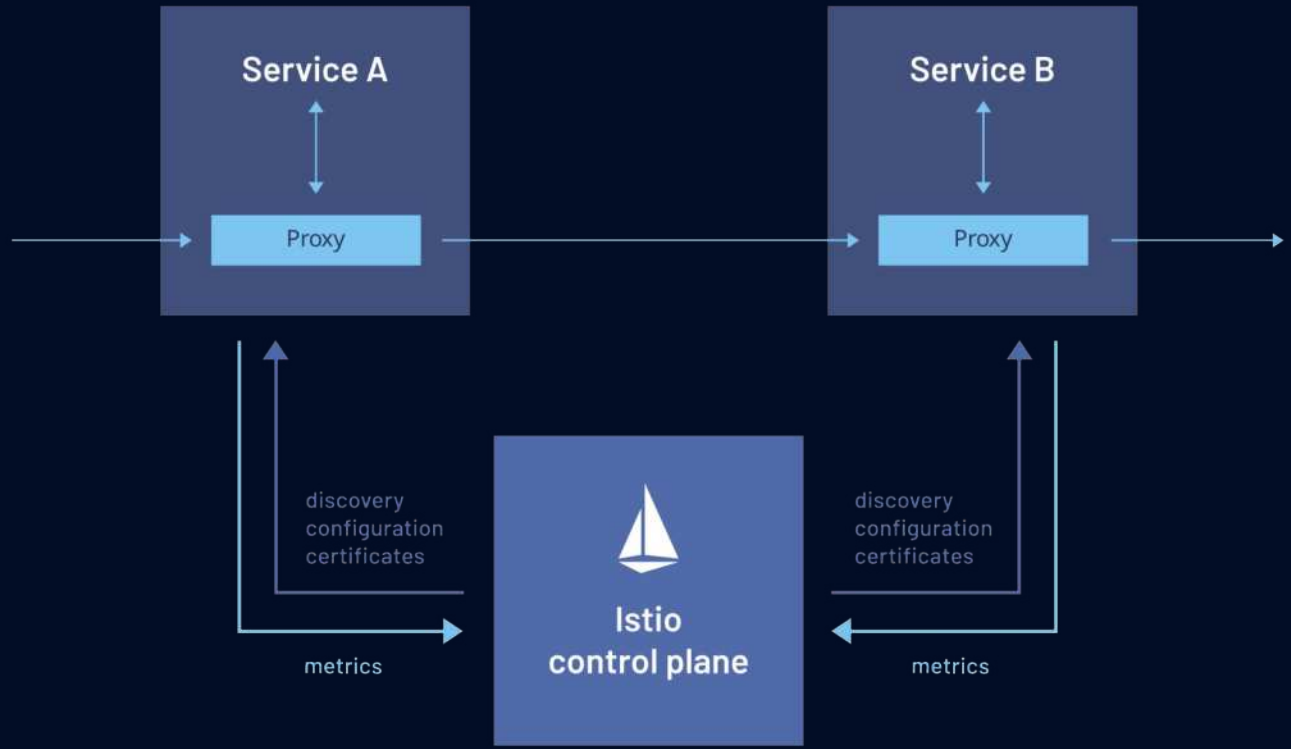
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What If We Just Catch All The Incoming And Exiting Traffic And Enforce SPIFFE/SPIRE?

That's (essentially) Envoy Proxy + Istio



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OK, but what about those of us not blessed to be 100% Cloud Native?



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WIMSE IETF Working Group

Workload Identity in Multi-System Environments

The IETF Workload Identity in Multi-System Environments (WIMSE) working group addresses authentication and identity challenges in distributed microservices architectures. WIMSE focuses on defining standardized solutions for managing workload identities and secure communication between services.

Key Objectives:

- Develop a unified architecture for securely managing workload identities.
- Create token-based solutions to authenticate and secure REST/HTTP communications between workloads.
- Ensure interoperability across diverse platforms.
- Collaborate with other IETF working groups and external organizations to ensure conhesive standards.

Most collaboration in the working group is via the mailing list.

For more information about IETF, visit the official IETF website.



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Workload Identity in a Multi System Environment (WIMSE) Architecture

- Workload

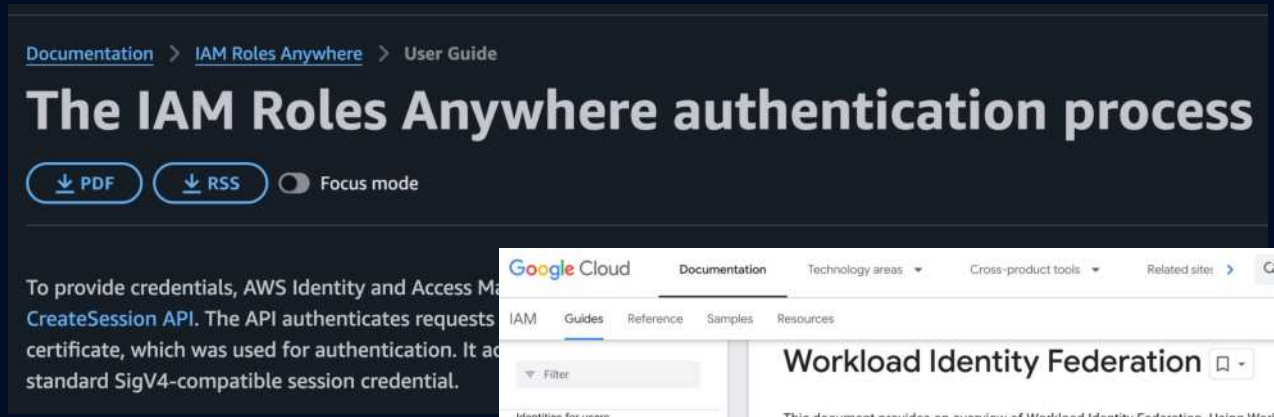
A workload is a running instance of software executing for a specific purpose. Workload typically interacts with other parts of a larger system. A workload may exist for a very short durations of time (fraction of a second) and run for a specific purpose such as to provide a response to an API request. Other kinds of workloads may execute for a very long duration, such as months or years. Examples include database services and machine learning training jobs.

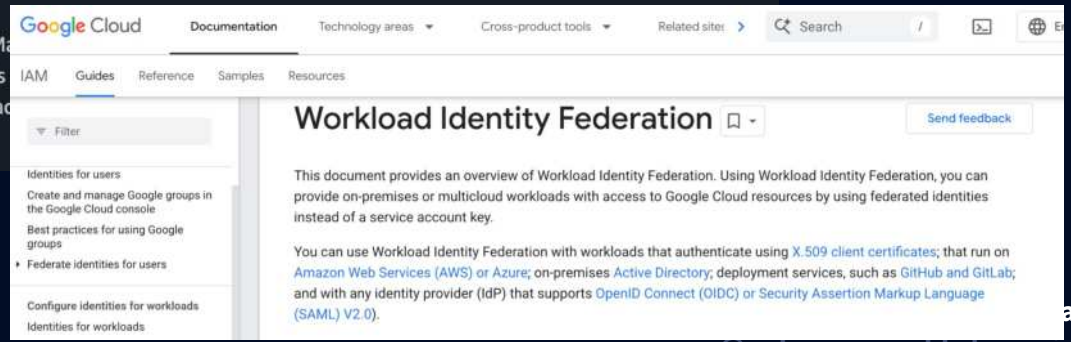
https://datatracker.ietf.org/doc/draft-ietf-wimse-arch/



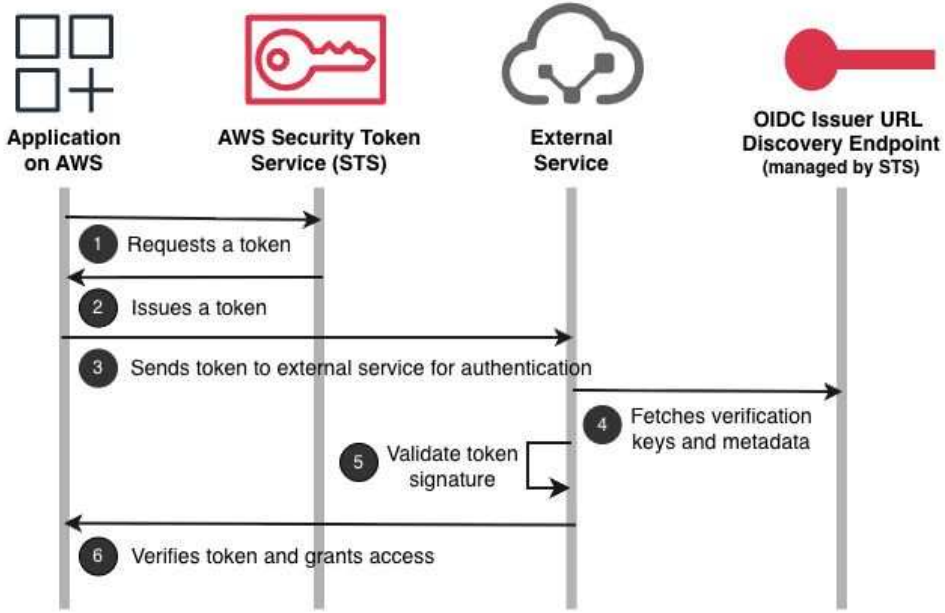
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Here's an example of getting an identity token using the AWS SDK for Python (Boto3). I can also do this using AWS Command Line Interface (AWS CLI).

Python

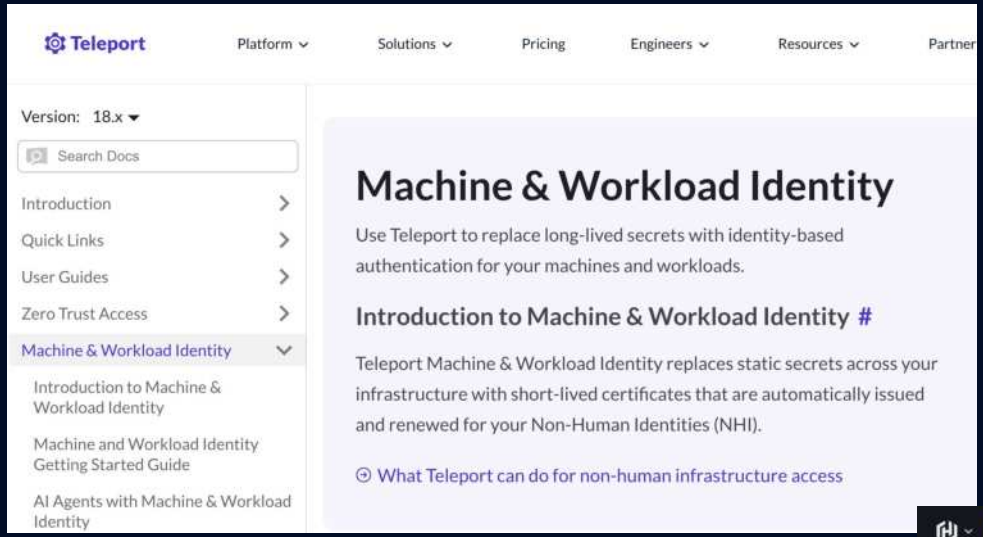
```python
import boto3

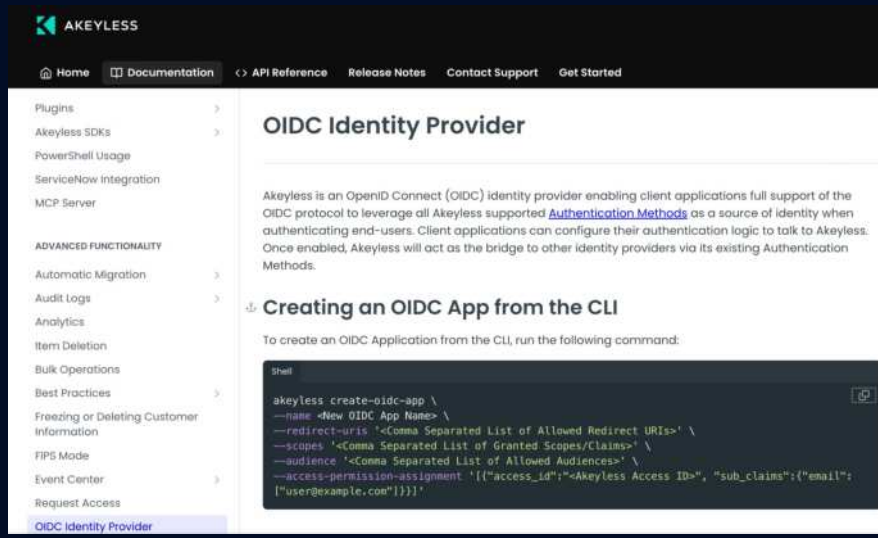
sts_client = boto3.client('sts')
response = sts_client.get_web_identity_token(
    Audience=['my-app'],
    SigningAlgorithm='ES384',  # or 'RS256'
    DurationSeconds=300
)
jwt_token = response['WebIdentityToken']
print(jwt_token)
```

- **Availability** – AWS IAM outbound identity federation is available at no additional cost in all AWS commercial Regions, AWS GovCloud (US) Regions, and China Regions.
- **Pricing** – This feature is available at no additional cost.

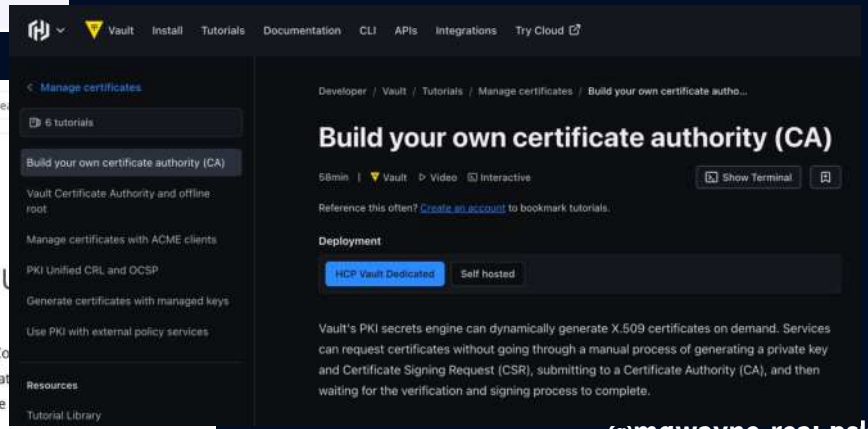


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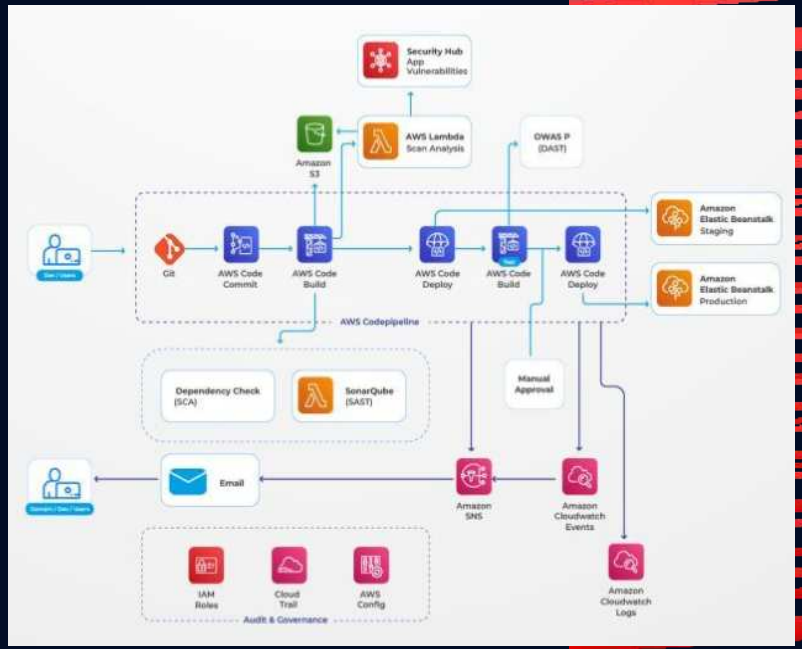








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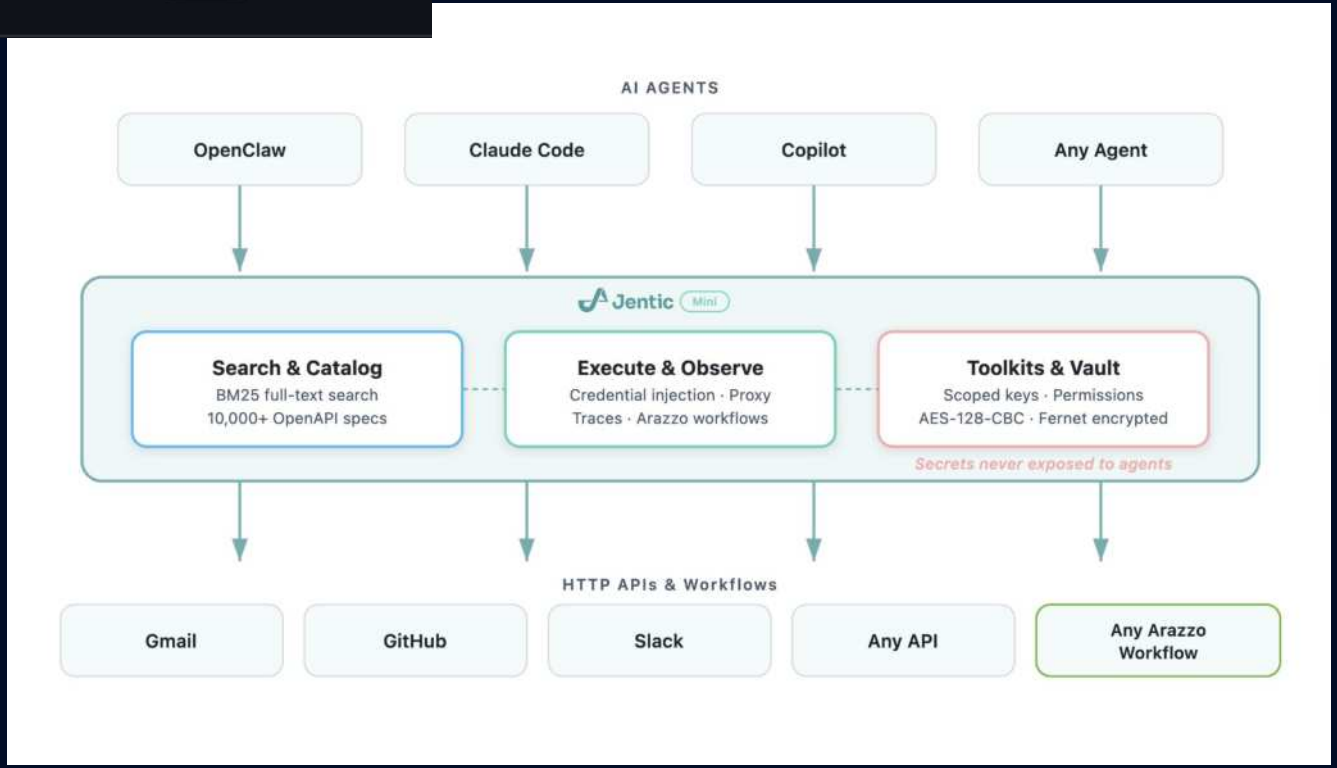
Non-human

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jentic / jentic-mini

Code Issues 40 Pull requ

jentic-mini Public



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May 21, 2026

Solving the Identity Crisis for AI Agents


Matt Mathew
Sr Staff Engineer

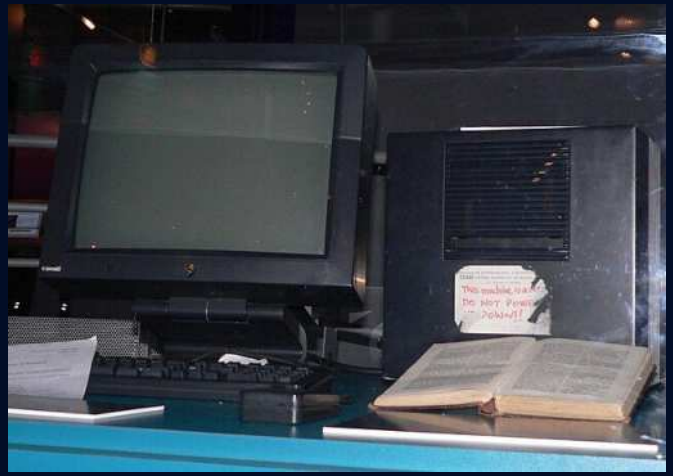

Prasad Borole
Staff Software Engineer


Meng Huang
Engineering Manager

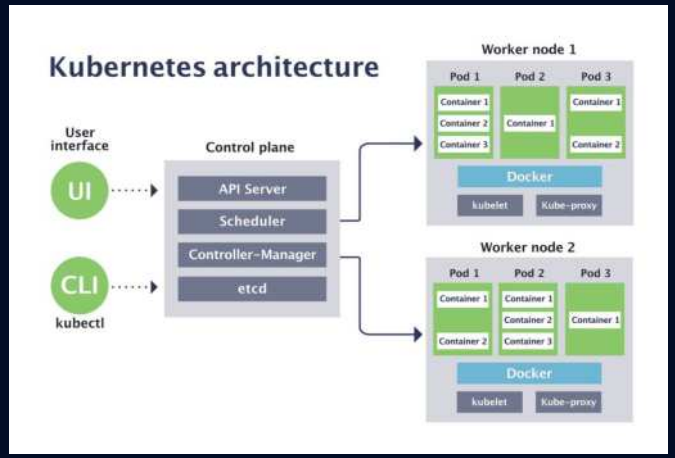
3+

https://www.uber.com/us/en/blog/solving-the-agent-identity-crisis/

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Hi. I'm Dwayne :



Dwayne McDaniel
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{

"Hometown" : "Chicago",

- "Mission" : "Help people figure stuff out",
- "Developer-advocate-since" : "2014",
- ~~"Host" : "The Security Repo Podcast",~~

"Co-Organizer" : "AppSec Village",

"Website" : "https://Dwayne-McDaniel.com",

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"mcdwayne@mastodon.social",

"https://ww.linkedin.com/in/dwaynemcdaniel" },

"Other-interests" : {"crochet", "karaoke",

"rock and roll concerts", "music in general"}

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About GitGuardian



GitGuardian is a secrets visibility platform helping teams solve Non-Human Identity security crisis



From Pets To Cattle To Agents: Evolving Identity And Security For Workloads

https://tinyurl.com/pets-sreday





“Step 1: Scope your work
The first step in the threat modeling process is concerned with gaining an understanding of what you’re working on.”

https://owasp.org/www-community/Threat_Modeling_Process

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GitGuardian

Yelp/**detect-secrets**





git-secrets



trufflesecurity/trufflehog

`git grep -E <pattern>`

PingSafe™



gitLeaks





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